Max Profit With K Transactions

AlgoExpert · Very Hard · Dynamic Programming

The problem

- Daily prices, at most k buy/sell pairs (no overlap); max profit.

Intuition

- For each transaction count, track the best profit by day using the best buy so far.

Approach

- 1. Rows for transactions, columns for days
- 2. Carry max of previous row minus price
- 3. Profit is max of skip or sell today
- 4. Two rows suffice

Java solution

```java
public class MaxProfitWithKTransactions {
    public static int maxProfitWithKTransactions(int[] prices, int k) {
        if (prices.length == 0) return 0;
        int[] prev = new int[prices.length], cur = new int[prices.length];
        for (int t = 1; t <= k; t++) {
            int bestBuy = Integer.MIN_VALUE;
            for (int d = 0; d < prices.length; d++) {
                bestBuy = Math.max(bestBuy, prev[d] - prices[d]);
                cur[d] = Math.max(d > 0 ? cur[d - 1] : 0, bestBuy + prices[d]);
            }
            int[] tmp = prev; prev = cur; cur = tmp;
        }
        return prev[prices.length - 1];
    }
}
```

Complexity

- Time: O(n*k)
- Space: O(n)

Recap

- For each transaction count, track the best profit by day using the best buy so far.
- Complexity: O(n*k)